Quasi-Poisson and Negative Binomial models

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Overdispersion

Overdispersion occurs when the response variable Y_i has greater variability than the model accounts for.

Poisson: $Var(Y_i|\mathbf{x}_i) = \mathbb{E}[Y_i|\mathbf{x}_i] = \mu_i$

If we fit a Poisson regression model, overdispersion means that $Var(Y_i|\mathbf{x}_i) > \mu_i$

quasi-Poisson models

$$\log(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$
$$Var(Y_i|\mathbf{x}_i) = \phi\mu_i$$

- Estimated coefficients ($\widehat{\boldsymbol{\beta}}$) are the same
- Different estimated variance-covariance matrix

Example: articles data

```
glm(art ~ phd + ment + kid5, family = poisson, data = artic
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.276493   0.087294   3.167  0.00154 **
phd          0.008071   0.026358   0.306  0.75944
ment         0.026537   0.002001  13.264  < 2e-16 ***
kid5        -0.099696   0.035248  -2.828  0.00468 **
(Dispersion parameter for poisson family
   taken to be 1)
```

```
glm(art ~ phd + ment + kid5, family = quasipoisson, data =
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *
(Dispersion parameter for quasipoisson family
   taken to be 1.88461)
```

Activity: comparing standard errors

Work on the class activity (course website).

Activity

Poisson output:

	Estimate	Std. Error	z value	Pr(>|z
(Intercept)	0.276493292	0.087294456	3.1673637	1.538278e-0
phd	0.008071036	0.026357684	0.3062119	7.594433e-0
ment	0.026537317	0.002000733	13.2637987	3.753884e-4
kid5	-0.099696463	0.035248105	-2.8284205	4.677832e-0

95% confidence interval for e^{β_3}:

Activity: comparing standard errors

Options for estimating $SE(\widehat{\beta}_j)$ when we fit a Poisson model:

- Estimate from the original Poisson model (may be wrong!)
- Bootstrap resampling (should be approximately correct even if model is misspecified)
- Sandwich variance (should be approximately correct even if model is misspecified)
- quasi-Poisson (inflate standard errors by $\sqrt{\widehat{\phi}}$)

Activity: comparing standard errors

- Poisson estimate: $\widehat{SE}(\widehat{\beta}_3)_{\text{poisson}} = 0.03525$
- Bootstrap estimate: $\widehat{SE}(\widehat{\beta}_3)_{\text{boot}} = 0.0490$
- Sandwich estimate: $\widehat{SE}(\widehat{\beta}_3)_{\text{sandwich}} = 0.0483$
- quasi-Poisson estimate: $\widehat{SE}(\widehat{\beta}_3)_{\text{quasi-poisson}} = 0.0484$

How do the estimates compare? What will be the effect (on confidence intervals and hypothesis tests) of incorrectly assuming a Poisson distribution when the data are overdispersed?

Negative Binomial models

$$Y \sim NB(\mu, r) \qquad f(y; \mu, r) = \frac{\Gamma(y + r)}{\Gamma(y + 1)\Gamma(r)} \left(\frac{\mu}{\mu + r}\right)^y \left(1 - \frac{\mu}{\mu + r}\right)^r$$

$$Y_i | \mathbf{x}_i \sim NB(\mu_i, r) \qquad \log(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

Example: articles data

```
m3 <- MASS::glm.nb(art ~ phd + ment + kid5, data = articles
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.233528   0.117585   1.986   0.0470 *
phd          0.008203   0.036069   0.227   0.8201
ment         0.030374   0.003240   9.373   <2e-16 ***
kid5        -0.093256   0.046475  -2.007   0.0448 *

(Dispersion parameter for Negative Binomial(2.1647)
   family taken to be 1)

AIC: 3145.6
```

Inference for negative binomial models

- Can use the usual Wald tests for β (from asymptotic normality)
- LRT also works because a full distribution is specified

Example: articles data

```
m3 <- MASS::glm.nb(art ~ phd + ment + kid5, data = articles
m4 <- MASS::glm.nb(art ~ phd + kid5, data = articles)

pchisq(m3$twologlik - m4$twologlik, 1, lower.tail=F)
```

```
## [1] 1.708206e-18
```